Lesson 2.1: Advanced data cleansing, verification and OCR

What you learn

- **Cleansing.** Trim spaces, fix case, standardise dates and phone numbers, remove duplicates.
- **Verification controls.** Validate fields, double-check key values and sample your own work.
- **OCR.** Convert scanned PDFs to text, then proofread; OCR makes errors on poor scans.

Tools and resources (official sites, not checked live)

- OpenRefine

ASSIGNMENT

Practice for lesson 2.1

Business problem: Validate records.

Steps

1. Run the tested validator.
2. Add two bad rows.
3. Report the error rate.

Expected output: A list of errors and a rate.

SOLUTION (ANSWER KEY)

PYTHON

```
import re

def clean_phone(raw):
    digits = re.sub(r"\D", "", raw)
    if digits.startswith("91") and len(digits) == 12:
        digits = digits[2:]
    return digits if len(digits) == 10 else None

def valid_email(raw):
    return bool(re.fullmatch(r"[^@\s]+@[^@\s]+\.[^@\s]+", raw.strip()))

def validate_record(rec):
    errors = []
    if not rec.get("name", "").strip():
        errors.append("name missing")
    if not valid_email(rec.get("email", "")):
        errors.append("email invalid")
    if clean_phone(rec.get("phone", "")) is None:
        errors.append("phone must be 10 digits (after removing +91)")
    return errors

def reconcile(bank, ledger):
    """Match bank lines to ledger entries by amount and date. Returns matched, unmatched_bank,
unmatched_ledger."""
    remaining = list(ledger)
    matched, unmatched_bank = [], []
    for b in bank:
```

```
        hit = next((l for l in remaining if l["amount"] == b["amount"] and l["date"] ==
b["date"]), None)
        if hit:
            matched.append((b["id"], hit["id"])); remaining.remove(hit)
        else:
            unmatched_bank.append(b["id"])
    return matched, unmatched_bank, [l["id"] for l in remaining]
```

Each bad record returns its problems, so errors are fixed from the source, not guessed.

Check yourself

1. Why proofread OCR output?
   Answer: OCR makes errors.
2. What is a verification control?
   Answer: A check that catches mistakes.

Lesson 2.1 of the **Admin and Business Support** lessons. See this lesson and its official links on skill navora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.